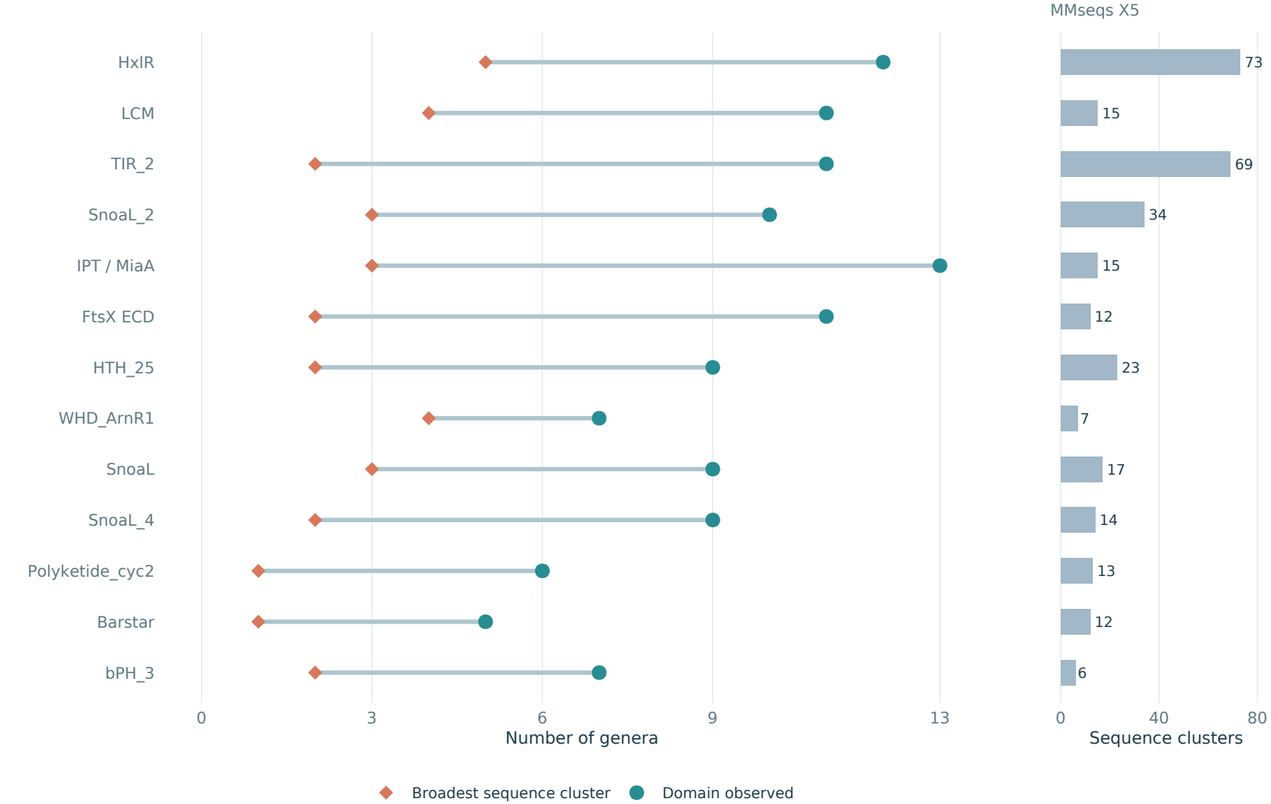
A

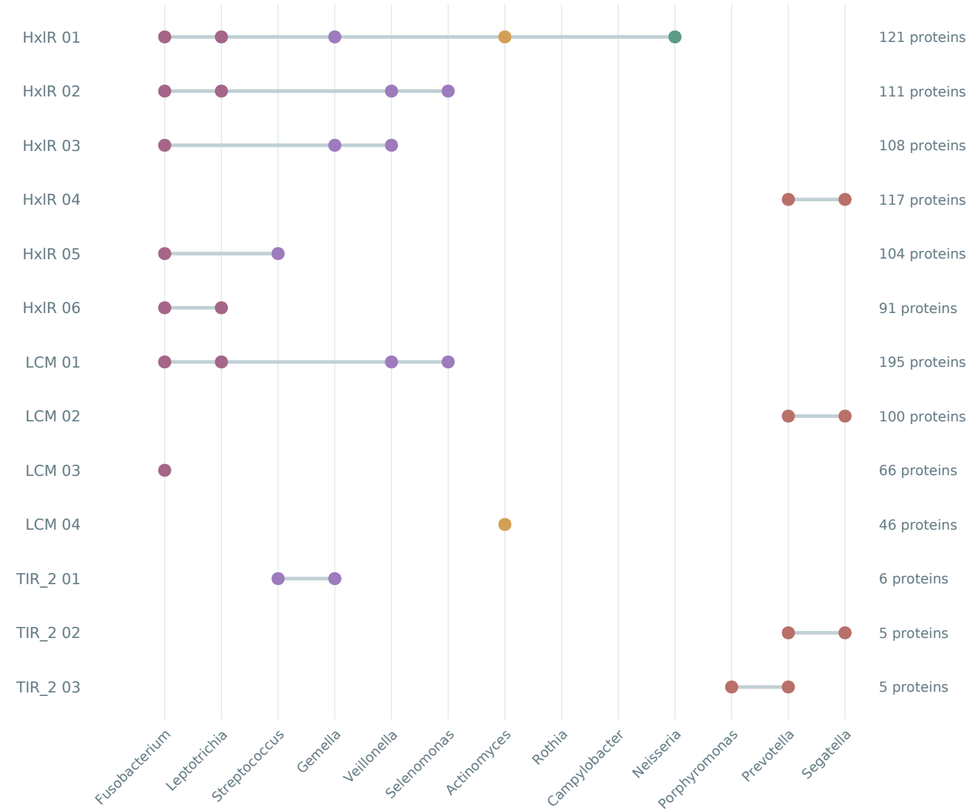
Domains encompass diverse protein families



B

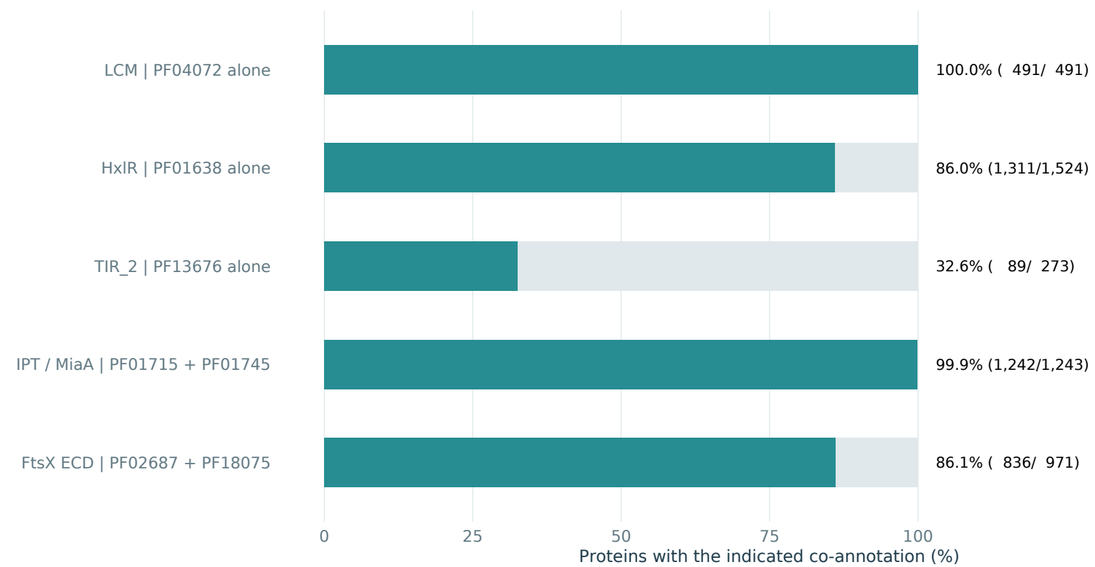
Selected protein clusters across genera

Rows are actual sequence clusters; coloured points show observed membership



C

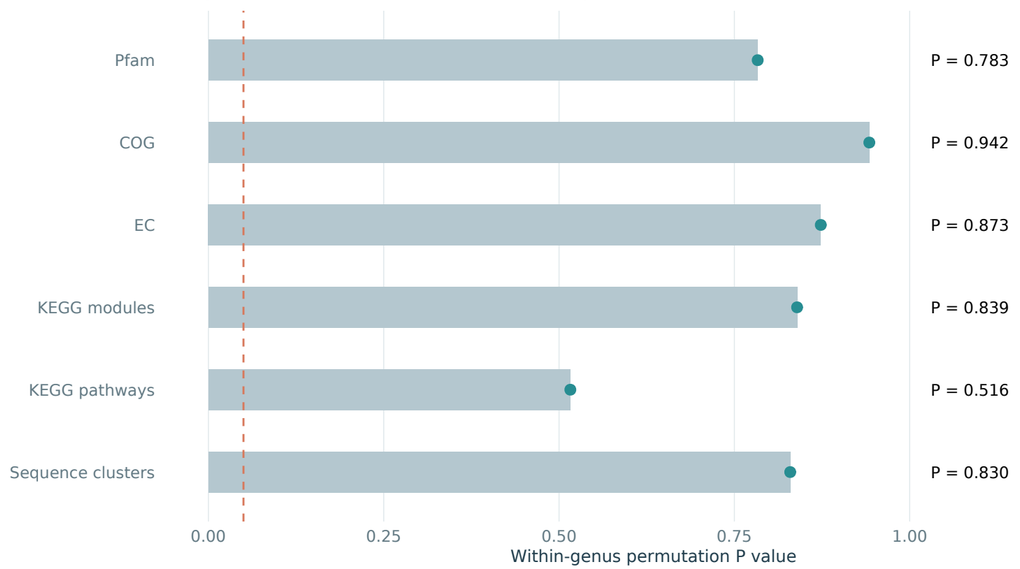
Dominant protein-domain combinations



D

Genome-wide directional coherence

No significant omnibus effect in the six annotation layers



Sequence clusters are not experimentally established orthologues. Genomes are nested within HMTs; neither domain co-annotation nor omnibus nonsignificance establishes biochemical mechanism.